

Learning from Click-Logs: a Two-Stage News Recommender for EB-NeRD and MIND

Design note, CS4.406 Information Retrieval & Extraction, Assignment 2

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Summary

A two-stage news recommender for EB-NeRD [1] and MIND [2]: cheap retrieval narrows the catalogue to 200 candidates, then a LightGBM LambdaRank model re-ranks them on 22 click-log features, all restricted to data from strictly before the impression predicted.

- **The behavioural axis is what matters.**

On EB-NeRD the re-ranker scores 0.7908 test AUC against 0.5380 for retrieval alone, a gain of +0.2528 [+0.2514, +0.2543]. Popularity carries most of it: a model given popularity and nothing else already reaches 0.7348. Because that signal needs no history, cold-start and warm users gain equally (0.7910 and 0.7908), but the re-ranker is *worse* than retrieval on the most popular articles.

- **The same architecture is worth far less on MIND**, +0.0583 [+0.0565, +0.0603] test,

because five of the 22 features have no source there: a measured property of the data, not a failure to reproduce.

- **We reproduced NRMS and beat it** with one change, a longer user history: +0.0094 [+0.0086, +0.0102] test AUC. Our re-ranker sits about 0.19 AUC above it, because NRMS never sees the click log’s popularity signal.

- **Honesty is nearly free.** Serving-unavailable features would add only +0.0226 [+0.0212, +0.0240], against the +0.2528 the behavioural axis earns legitimately. We found and fixed a future-click leak contaminating 99.8% of validation impressions; fixing it *raised* test accuracy.

- **Serving fits the budget, and exact search breaks at scale.** A single request takes 42.5 ms at the median and 77.3 ms at p99, inside a 100 ms target. At 10× the catalogue exact embedding search grows 26.8×

and becomes the bottleneck; the candidate pool stays flat at 1.13×, so the other stages do the same work more slowly.

- **What we upload is not what we report**, and we say so: Codabench withholds the click labels, so the submitted model drops five popularity features and scores 0.7351 against 0.7908 on our own test split.

1 What we built

The unit of everything is an *impression*: at time t a user is shown *candidate* articles and clicks some. Our job is to order each impression’s candidates so the clicked ones come first. We extend our two Assignment-1 retrieval systems, lexical and semantic, with the *behavioural* signal in the logs: what users clicked, how often each article was clicked or shown, and how fresh it is.

1.1 Data and evaluation protocol

Splits are by time, never random, since a random split lets the model learn from Thursday to predict Wednesday. EB-NeRD val begins 23 May 07:00 and test 25 May 07:00; MIND uses the last day of its training log as val and its dev log as test.

Metric: per-impression AUC, averaged over impressions containing both a click and a non-click, never pooled across impressions. **Significance: paired bootstrap 95% intervals**, 1,000 resamples on shared impressions; a gain counts only if its interval excludes zero. Selection happens on val and test is scored once. Both choices, and what pooling would have cost, are defined in Appendix A.

dataset	articles	train / val / test impressions	leaderboard test set	
			impressions	articles
EB-NeRD small (Danish)	20,738	168,522 / 64,365 / 244,647	13,536,710	125,500
MIND small (English)	65,238	95,071 / 61,894 / 73,152	2,370,727	120,961

Table 1: Development data and the held-out leaderboard sets. Splits are temporal: every train impression precedes every val impression, which precedes every test impression.

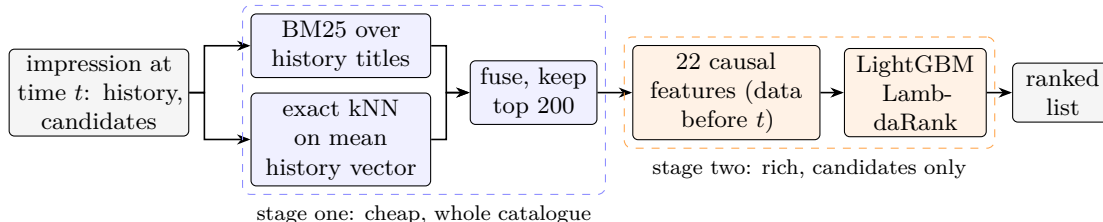


Figure 1: The two-stage pipeline. Stage one touches every article, so it must stay cheap per article. Stage two touches only the candidates, so it can afford per-candidate features.

1.2 Architecture: retrieve, then re-rank

Stage one, candidate generation (carried from Assignment 1). *BM25* [4, 5] over a query built from the titles of the user’s last N clicks ($N = 30$ on EB-NeRD, 100 on MIND), stemming on for both languages, Danish stopword list. The *semantic* retriever takes the mean of the user’s clicked-article vectors and finds nearest neighbours by cosine with an exact FAISS IndexFlatIP [6]. Vectors are the provided 768-dimensional `contrastive_vector` for EB-NeRD and 384-dimensional `all-MiniLM-L6-v2` [7] for MIND, both chosen by measurement in Assignment 1. The two scores are standardised within each pool and mixed linearly, weight chosen on val. Each retriever keeps its top 200. Both indexes are held in RAM.

Stage two, the re-ranker. LightGBM [8] with the *LambdaRank* objective [9], measured against a binary objective rather than assumed (Table 4). Settings: learning rate 0.05, 31 leaves, at least 50 rows per leaf, 90% feature and row subsampling, up to 400 rounds with early stopping on val nDCG, seed 13. Appendix A defines both retrieval scores and the objective.

1.3 Features and the behaviour-window boundary

Every feature for an impression at time t uses only data from strictly before t , the *behaviour-window boundary*. Click and exposure counts come from binary search over each article’s sorted event times with a strict “ $< t$ ” cut, so an event at exactly t is excluded. Tests assert this on every split and each was checked by injection (Section 2.5).

Five features cannot exist on MIND, and we drop them explicitly. Its publication time is null for all 65,238 articles and it ships no read-time or scroll data. Since a column of zeros looks like a real feature to LightGBM, the builder refuses to emit anything all-null or constant. This matches our pre-design data audit: EB-NeRD is rich in behavioural signal, MIND has little beyond the click itself.

Exposure (“how often was this article shown before t ”) was added because the leaderboard files withhold clicks (Section 4) but keep the in-view lists, and kept for a better reason: a live recommender knows its own exposure log immediately, while click feedback arrives later.

family	features and what they measure	EB-NeRD	MIND
stage-one scores	bm25 , emb (the two retrieval scores); emb_rank (the embedding score’s rank inside the impression)	yes	yes
click popularity	pop_causal : clicks before t in a 6 h window (unbounded on MIND); pop_24h : the same over 24 h; pop_velocity : their ratio, “rising now” versus “steadily popular”; pop_rank , pop_rel_max : standing among this impression’s candidates	yes	yes
exposure	the same five shapes (exp_causal , exp_24h , exp_velocity , exp_rank , exp_rel_max) counting how often an article was <i>shown</i> before t	yes	yes
user match	cat_match : share of the user’s history in the candidate’s category; ent_overlap : shared named entities with the history	yes	yes
context	hist_len : history length; n_cands : number of candidates in the impression	yes	yes
freshness	age_hours , recency (24 h half-life decay of age)	yes	no
engagement	user_read , user_scroll : mean read time and scroll depth of past clicks; engage_sim : cosine to a read-time-weighted user vector	yes	no
total		22	17

Table 2: The re-ranker’s features. Rank and relative features exist because the metric is decided inside one impression: a popularity of 200 dominates a quiet impression and is ordinary in a busy one.

2 Results

Unless stated otherwise, numbers are per-impression AUC on EB-NeRD small and MIND small, with paired bootstrap 95% intervals, selected on val and scored once on test.

2.1 Before and after re-ranking

Two baselines appear in this report and they are not the same thing. Here “stage one” is the fused retrieval score ranking the candidates directly (0.5380 on test). The ablation grid in Table 4 instead compares against **stage1_only**, a ranker trained on the two retrieval scores alone, which is slightly stronger at 0.5430. That is why the same result reads +0.2528 here and +0.2223 there.

On EB-NeRD the re-ranker is worth +0.2528 AUC over fused retrieval, [+0.2514, +0.2543] on test, and +0.2357 [+0.2329, +0.2385] on val. The val and test intervals overlap, so the gain generalises across the time boundary rather than being an artefact of what we selected on. Every other accuracy metric moves the same way, so this is a genuine reordering rather than a trade between metrics.

On MIND the same architecture is worth +0.0583 [+0.0565, +0.0603], far less. Two things compound: five of the 22 features have no source there (Section 1), so the behavioural axis is thinner, and MIND’s stage one is already stronger in absolute terms (0.6381 against EB-NeRD’s 0.5380), leaving less headroom. A smaller gain on a dataset with less signal is a measured property of the data, so we report the contrast rather than presenting one system that is quietly two.

One family of features did transfer, and it was the one built under duress. An earlier round of seven features moved MIND by nothing significant at all. The five exposure features, added because Codabench withholds the click labels, are worth +0.0297 [+0.0278, +0.0316] on MIND test and +0.0458 [+0.0440, +0.0478] on val. How often an article was *shown* is apparently a signal both corpora carry, where freshness and engagement are specific to EB-NeRD.

system	AUC	MRR	MRR, all clicks	nDCG@5	nDCG@10
<i>EB-NeRD small, test</i>					
BM25 (stage one)	0.5107 [0.5094, 0.5120]	0.3257 [0.3245, 0.3269]	0.3253 [0.3241, 0.3264]	0.3577 [0.3563, 0.3591]	0.4409 [0.4397, 0.4420]
embeddings (stage one)	0.5397 [0.5384, 0.5409]	0.3492 [0.3480, 0.3503]	0.3487 [0.3475, 0.3498]	0.3831 [0.3818, 0.3845]	0.4627 [0.4616, 0.4638]
fused (stage one)	0.5380 [0.5367, 0.5392]	0.3474 [0.3463, 0.3486]	0.3470 [0.3458, 0.3481]	0.3813 [0.3800, 0.3827]	0.4613 [0.4602, 0.4624]
two-stage re-ranker	0.7908 [0.7898, 0.7918]	0.5685 [0.5671, 0.5699]	0.5678 [0.5665, 0.5693]	0.6330 [0.6318, 0.6344]	0.6650 [0.6639, 0.6662]
fused + lifetime popularity*	0.5797 [0.5785, 0.5810]	0.3688 [0.3677, 0.3700]	0.3684 [0.3672, 0.3696]	0.4101 [0.4088, 0.4115]	0.4841 [0.4830, 0.4852]
<i>MIND small, test</i>					
BM25 (stage one)	0.5685 [0.5663, 0.5706]	0.3108 [0.3085, 0.3133]	0.2693 [0.2673, 0.2717]	0.2868 [0.2844, 0.2894]	0.3479 [0.3456, 0.3504]
embeddings (stage one)	0.6369 [0.6347, 0.6390]	0.3510 [0.3486, 0.3535]	0.3061 [0.3039, 0.3085]	0.3341 [0.3316, 0.3367]	0.3937 [0.3913, 0.3961]
fused (stage one)	0.6381 [0.6358, 0.6402]	0.3536 [0.3511, 0.3561]	0.3075 [0.3053, 0.3098]	0.3361 [0.3335, 0.3388]	0.3956 [0.3934, 0.3981]
two-stage re-ranker	0.6964 [0.6943, 0.6983]	0.4027 [0.4001, 0.4054]	0.3459 [0.3436, 0.3483]	0.3854 [0.3827, 0.3882]	0.4458 [0.4435, 0.4483]

Table 3: All accuracy metrics on the test split, with 95% bootstrap intervals beneath each mean. “Stage one” rows are the retrieval scores ranking the same candidates; the re-ranker reorders them. *Lifetime popularity is not available at serving time and is reported only to price what it would buy (Section 2.5). Validation-split figures are in Appendix I.

2.2 What each design choice was worth

Every choice below was made by measuring the alternative, one change at a time, with a paired bootstrap interval. Rows marked “10 features” were measured before the feature set grew; the two feature rows are the incremental steps to the shipped EB-NeRD model.

Three deserve a sentence. **The objective matters:** a binary objective leans on absolute freshness, while LambdaRank leans on popularity contrasts inside the impression, which is what decides the ranking. **Recent popularity beats lifetime popularity:** counting only the last 6 hours asks “how popular is this right now”. In isolation that moved the feature from 0.6902 to 0.7349, but inside the ranker the gain is far smaller because other features cover for it, so we quote the ranker figure. **Fusion helps on val and hurts on test**, both significantly: the mixing weight does not transfer across the time boundary. That is moot next to stage two, but it is an honest negative result.

2.3 Baseline reproduced, then beaten (EB-NeRD)

The baseline. NRMS [3] (Appendix A), run from the organisers’ `ebnerd_nrms_docvec.py` unmodified on EB-NeRD small with the provided document vectors, history size 20, 4 negatives per positive, batch 32, seed 16: val AUC **0.6484** (best epoch 3, early-stopped at 7, 54 minutes on 16 CPU cores of a cluster node).

That number is not comparable to ours. The benchmark script trains on the organisers’ train and validation blocks together and validates on its own last day, which in our split means training on our val and test impressions. So we drove the same model, settings, vectors and loader from our temporal split instead: train on our train, stop early on our val, score test once. That run is the baseline below.

The one principled change: a longer history, 20 to 50 clicks. The motivation is our own measurement: on clean data the mean-of-history user vector improves steadily with history, 0.5416 at 10 clicks to 0.5545 at 100 (Appendix H), and NRMS builds its user representation from exactly that history. It does benefit, significantly on both splits and more on

choice	compared against	val Δ [95% CI]	test Δ [95% CI]	verdict
re-ranking at all	stage one alone	+0.2138 [+0.2110, +0.2168]	+0.2223 [+0.2209, +0.2238]	core result
LambdaRank objective (10 features)	binary objective	+0.0161 [+0.0150, +0.0173]	+0.0165 [+0.0159, +0.0171]	kept
6 h click-popularity window (10 features)	all past clicks	+0.0034 [+0.0026, +0.0043]	+0.0062 [+0.0058, +0.0067]	shipped
seven rank, velocity and context features	the 10 features	+0.0113 [+0.0104, +0.0122]	+0.0130 [+0.0124, +0.0134]	shipped
five exposure features	the 17 features	+0.0249 [+0.0238, +0.0261]	+0.0255 [+0.0249, +0.0261]	shipped
read-time-weighted user vector	uniform vector only	+0.0003 [−0.0003, +0.0010]	+0.0005 [+0.0002, +0.0008]	kept, tiny
linear fusion of BM25 and embeddings	embeddings alone	+0.0022 [+0.0010, +0.0034]	−0.0017 [−0.0023, −0.0010]	does not generalise
title-only BM25 documents	title and abstract	+0.0025 [+0.0002, +0.0047]	+0.0043 [+0.0031, +0.0055]	not shipped [†]
stemming off, MIND	stemming on	−0.0023 [−0.0034, −0.0013]	−0.0028 [−0.0039, −0.0018]	keep on
stemming off, EB-NeRD	stemming on	−0.0003 [−0.0019, +0.0014]	−0.0002 [−0.0011, +0.0007]	keep on, for cost

Table 4: Per-impression AUC change from each choice, on EB-NeRD unless marked. The first row is the **stage1_only** arm of the 17-feature grid. [†]Title-only documents also cost val recall@200 by −0.0024 [−0.0035, −0.0013], and recall is what stage two consumes.

EB-NeRD small, our temporal split	val AUC	test AUC
NRMS, benchmark script and protocol (not comparable)	0.6484	—
NRMS baseline, history size 20	0.5969	0.5938
NRMS improved, history size 50	0.6042	0.6032
improved — baseline	+0.0072 [+0.0058, +0.0088]	+0.0094 [+0.0086, +0.0102]
our two-stage re-ranker	0.7885	0.7908

Table 5: NRMS reproduced and beaten with one change, paired bootstrap. Both NRMS arms score the identical 64,365 val and 244,647 test impressions, aligned by impression id.

test than val, the opposite of what over-fitting to val would produce. We stopped at 50 because attention cost grows with the square of history length. A guard asserts every article id reaches the model (coverage 1.0000 on all four checks), since a silent id mismatch would train on zero vectors and still report a believable AUC.

Why our system beats NRMS by about 0.19 AUC. Not because trees beat attention: NRMS reads article content and click order only, with no click popularity, our most valuable signal. Our own content-only arm **stage1_only** scores 0.5498 val, close to NRMS. The compar-

ison is between a model that sees the click log and one that does not.

2.4 Slices and beyond-accuracy

An average can hide a system that fails a subgroup, so we split test impressions by user history length and by how often the clicked article was clicked in training. **Slice sizes are printed because the thresholds are not obvious:** over 90% of articles in both datasets have zero training clicks, so a percentile over the whole catalogue would be zero and would call everything head. Taken over articles seen in training it gives

a head threshold of 349 clicks on EB-NeRD, 803 impressions, 0.3% of the split.

The gain is uniform across the slices that matter, with cold, warm and tail all improving by about 0.25 AUC. **Cold and warm users are now indistinguishable** (0.7910 and 0.7908), which is a property rather than an achievement: the model’s strongest features are popularity and exposure, which need no history, so having little history costs almost nothing.

Head articles are the exception, and the result reverses across the split boundary.

On validation the re-ranker beats fused retrieval on head (0.7080 [0.6905, 0.7253] against 0.6638 [0.6467, 0.6800]); on test it loses (0.5530 [0.5316, 0.5728] against 0.6353 [0.6143, 0.6574]). Both intervals are disjoint, but an effect that flips sign across a temporal boundary is a property of the window, not a stable claim. **So we cannot claim the re-ranker helps on the most popular articles, and on test it is clearly worse there than retrieval alone.** One untested explanation fits: our popularity counts only the last six hours and is rank-normalised inside the impression, a deliberate “trending now” signal that discounts steadily popular articles. Consistent with that, the leaky lifetime-popularity arm is the best system on head (0.6921), the only slice it wins, and MIND, whose window is unbounded, goes the other way: 0.7884 against 0.6347 on its 860 head impressions, its largest slice gain. MIND’s users also gain most where they have least, 0.6288 against 0.5125 on 2,214 impressions with no history at all.

The accuracy gain is not paid for in diversity or novelty (Appendix F). The re-ranker has the highest novelty of any system on both datasets and diversity close to BM25’s, so it is not collapsing onto a few popular stories. It does cost a little catalogue coverage: 0.2051 against 0.2026 on EB-NeRD test and 0.0529 against 0.0494 on MIND, roughly 3 to 7% fewer distinct articles ever surfacing. That is the real trade and it is small.

Coverage is quoted without an interval, on purpose, because a bootstrap systematically undercounts a distinct-item statistic. Our own

failed attempt is the demonstration and is in Appendix F.

2.5 Anti-gaming: serving-unavailable features and the leak we found

What the system declines to use. EB-NeRD ships three lifetime totals per article (views, page views, read time) summed over the whole collection period, so for any impression they include the future and cannot exist when serving. We trained the same ranker on the same rows with and without them:

In Assignment 1 the same totals were worth +0.042 to +0.075, more than our whole semantic axis, because they competed against systems with no popularity signal at all. Here they compete with click counts restricted to the past, which capture most of the same information legitimately. Cheating would buy +0.0226; the behavioural axis earns +0.2528 legitimately, roughly an order of magnitude more. Both arms are the 10-feature model, so this prices the leak against that round rather than the shipped 22. End to end the servable re-ranker (0.7908) far outscores stage one with lifetime popularity added (0.5797). MIND cannot run this comparison at all: its lifetime columns are null for all 65,238 articles.

Enforcement. Two tests guard the boundary: one asserts that no click, exposure or engagement value used for an impression comes from at or after its timestamp, on every split; the other fails if any scoring module reads a lifetime column. Each was verified by injecting a violation and watching it fail, after we found checks that passed without checking anything (Section 5).

The first recounts popularity against the **6 h window the shipped feature actually uses**, with a *two-sided* mask and a different code path from the builder’s binary search. Two-sided is load-bearing: asserting only “no more than the clicks before t ” is satisfied by a window reaching past t so long as it stays narrow, so the weaker check passes on the leak it exists to catch. We confirmed the stronger one is live by mutation rather than argument, since changing the window to 12 h or unbounded without rebuilding

slice	impressions	fused (stage one)	two-stage re-ranker	gain	fused + lifetime pop.*
<i>EB-NeRD small, test</i> cold: history ≤ 34 clicks; head: clicked article with ≥ 349 train clicks					
cold	25,105	0.5449 [0.5407, 0.5488]	0.7910 [0.7879, 0.7942]	+0.2461	0.5906 [0.5865, 0.5945]
warm	219,542	0.5372 [0.5358, 0.5386]	0.7908 [0.7897, 0.7917]	+0.2535	0.5785 [0.5771, 0.5798]
head	803	0.6353 [0.6143, 0.6574]	0.5530 [0.5316, 0.5728]	-0.0823	0.6921 [0.6700, 0.7129]
tail	243,844	0.5377 [0.5364, 0.5390]	0.7916 [0.7906, 0.7925]	+0.2539	0.5793 [0.5781, 0.5806]
<i>MIND small, test</i> cold: history ≤ 3 clicks; head: clicked article with ≥ 152 train clicks					
cold	7,529	0.5601 [0.5533, 0.5676]	0.6480 [0.6415, 0.6548]	+0.0880	-
warm	65,623	0.6470 [0.6448, 0.6492]	0.7019 [0.6999, 0.7039]	+0.0549	-
zero_history	2,214	0.5125 [0.5007, 0.5246]	0.6288 [0.6161, 0.6404]	+0.1163	-
head	860	0.6347 [0.6167, 0.6508]	0.7884 [0.7783, 0.7984]	+0.1538	-
tail	72,292	0.6381 [0.6359, 0.6401]	0.6953 [0.6932, 0.6975]	+0.0571	-

Table 6: Per-impression AUC by slice, test split, with 95% intervals. “Gain” is the re-ranker minus fused retrieval. *Lifetime popularity, not servable, shown for comparison.

EB-NeRD small, val, 64,365 impressions	per-impression AUC
causal features only (servable)	0.7523
plus lifetime totals (not servable)	0.7749
difference, paired bootstrap	+0.0226 [+0.0212, +0.0240]

Table 7: Metrics with and without serving-unavailable features, measured on the 10-feature model with the 6 h window.

makes both boundary tests fail. The suite is **29 passed, 6 skipped, 0 failed**.

We also found a real leak in our own code, and fixing it raised accuracy. Merging EB-NeRD’s two history blocks described **99.8% of validation and 99.9% of training impressions with history from their own future**; test was clean at 0.0%. It was worth +0.0967 on the affected feature, and after the fix test AUC rose from 0.7429 to 0.7461, because test features were always clean and only the model changed. How we found it, and the evidence that the fix was right, are in Appendix B.

3 Serving and scale

Every number here came from one machine, a laptop with 8 physical (12 logical) cores, 15.3 GB RAM and no GPU, timing the shipped 22-feature model with its 399 trees; none are mixed

across machines. Requests are timed **one at a time**, never batched, because batching raises throughput while making each request slower and it is the tail a user notices.

3.1 Index memory, in RAM and on disk

The dense index is the largest structure despite covering the fewest objects: a vector spends memory on every dimension of every article, while an inverted index spends it only on terms that occur. It is also the only one whose RAM and disk sizes agree, because a flat index *is* its raw matrix. Both facts matter for the scaling result.

3.2 Latency of a single request

The median request takes 42.5 ms and p99 77.3 ms. Exact embedding search is 44% of the median and the re-ranker 39%, so those two are

index	size	RAM (MiB)	disk (MiB)	RAM / disk
BM25 (sparse matrix)	297,145 postings, 30,388 terms	7.0	3.0	2.32×
FAISS flat index	20,738 × 768 float32	60.8	60.8	1.00×
feature store	5,514,689 candidate rows	547.0	240.4	2.28×

Table 8: Footprints measured separately in memory and serialised, since the two differ by different factors per structure.

stage	p50 (ms)	p95 (ms)	p99 (ms)	share of p50
tokenise query	1.49	2.36	5.31	4%
BM25 top 200	3.82	9.36	12.41	9%
embedding top 200 (exact)	18.65	35.54	48.08	44%
assemble features	0.60	0.71	2.10	1%
LightGBM predict	16.37	26.85	34.72	39%
total	42.45	64.66	77.31	

Table 9: End-to-end latency over 2,000 single requests on EB-NeRD small, stage by stage.

what to attack.

The accuracy gain was not free. The same bench on the earlier 10-feature model with 311 trees gave a median of 22.06 ms and p99 64.90 ms, so going to 22 features roughly doubled the median. That is the cost of the +0.2528 AUC, and it only appears when serving is measured rather than assumed.

3.3 Cost per 1000 queries at a target SLA

Against a target of p99 under 100 ms, the measured 77.3 ms **meets the SLA with 1.3× headroom**, down from the 1.5× the smaller model had. One core sustains 24 QPS, measured. Scaling to 8 physical cores gives 188 QPS and, at an assumed \$0.040 per vCPU-hour, **\$0.000472 per 1000 queries**. Both of those are *projections*: only the latency they multiply is measured. We count physical cores because this path is dense floating-point work, where a second thread on the same core shares an execution unit.

3.4 What breaks first at 10×

Extrapolating one measurement would assume the growth law that is exactly the question, so we subsampled the corpus and re-timed every stage.

Exact embedding search breaks first. Over

a 10.0× corpus it grows 26.8×, where a flat scan should be at worst linear. The cause is the memory hierarchy, not arithmetic: at the smallest corpus the vectors occupy 6.1 MiB and sit in cache, and at full size they occupy 60.8 MiB and every query streams them from RAM. **At 10× the catalogue this is a memory-bandwidth problem, not a FLOPs problem**, so buying cores would not fix it.

The re-ranker also slows down, and it is not doing more work. Its cost grew 7.6×, surprising because it scores only the candidate pool, whose size the retrievers set rather than the corpus. We measured the pool rather than assuming: across the same 10.0× corpus the mean pool goes 345 to 388 candidates, a factor of 1.13. The work is flat, so the extra time is the same work executing more slowly. Every stage slows at the largest corpus, including feature assembly, which touches only that pool. The consistent reading is that the dense scan’s memory traffic evicts everything else from cache, so the index taxes the whole request rather than only its own stage.

Two growth figures are upper bounds. Subsampling also shortens the query, which is built from history-article titles: mean query length falls 144.6 to 17.7 tokens and resolvable history items 29.1 to 3.6. So the 4.6× for tokenising and 16.2× for BM25 mix corpus size with query

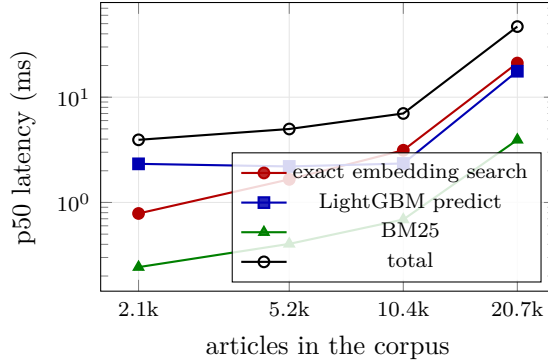


Figure 2: Per-stage median latency against corpus size, both axes logarithmic. The figures behind it are in Appendix G.

length and overstate the corpus effect. Exact search is unaffected, since a flat scan costs vectors times dimensions whatever the query holds, which is why the verdict rests on it.

This overturns our Assignment-1 conclusion that the per-impression Python loop would break before the linear algebra. The lever now is an approximate index, which Assignment 1 measured as 10 to 20 $\times$ faster and set aside as unnecessary, flagging a revisit at 10 $\times$. This is that revisit and the answer is the opposite. It would speed up more than its own stage, since it is also what taxes the others, but what it costs in recall here we have not measured. Repeated runs on this machine differ by roughly $\pm 20\%$.

4 Leaderboard submissions

The uploaded system is a re-ranker, but not the one reported above, and the gap is not in our favour. Both are stated so that the better number never stands in for the uploaded one.

Why the two differ. The Codabench test files withhold `article_ids_clicked`, checked by comparing columns rather than assumed: training behaviours minus test behaviours leaves exactly the outcome columns. Our five click-popularity features count clicks before each impression, and on the test period those clicks *are* the withheld labels, so they cannot be computed there at any cost; the five exposure features replace them, needing only the in-view lists the files do contain. We also dropped `bm25`, for cost

rather than availability: it scores every article for every distinct history, about 1.6 hours on MIND-large, and is worth +0.0004 test AUC.

Scale and validation. The EB-NeRD test set has 13,536,710 impressions and 205,925,868 candidate pairs; MIND-large has 2,370,727. Neither fits in memory, so prediction streams in slices. Scoring EB-NeRD took 3 h 06 m at 44 GB peak on a cluster node, after three earlier implementations were abandoned once their measured rates projected 38, 90 and 135 hours; batching the LightGBM call, 13.5M invocations down to 541, was the fix that mattered, and each rewrite was checked for byte-identical output. Because MIND allows one submission per day, every file passes an offline validator first: exact inner file-name, nothing else in the zip, row count, row order, and each row a permutation of $1..n$. Row order matters most, since ranks are positional and a reordered file scores as a plausible bad model rather than failing. The validator was itself checked by injecting five violations.

The MIND upload is marginally below our Assignment 1 entry, and that is the honest result. It is not a regression in the re-ranker. MIND ships almost no behavioural signal to begin with (Section 1), so the submitted system there is 11 features against EB-NeRD’s 22, and the click-popularity family carrying most of the gain cannot be computed on the test period. What remains is close to the Assignment 1 system plus exposure, and lands within 0.003 of it. The behavioural axis is worth +0.2077 AUC

EB-NeRD small, our test split	features	per-impression AUC
reported system	22	0.7908
uploaded system (no click popularity, no <code>bm25</code>)	16	0.7351
uploaded – reported, paired bootstrap		−0.0556 [−0.0565, −0.0547]

Table 10: The gap between what we report and what we upload, measured on our own labelled test split. The MIND upload uses 11 features and scores 0.6460 on our MIND test split.

leaderboard	AUC	MRR	nDCG@5	nDCG@10
MIND (Codabench 13967), submission 932812	0.6473	0.3174	0.3448	0.4010
our A1 entry, for comparison	0.6503	0.3198	0.3454	0.4010
EB-NeRD (Codabench 2469)	[not scored]	[—]	[—]	[—]

Table 11: Official leaderboard scores. MIND scored on 2026-09-19 at rank 54; screenshot in Appendix K. EB-NeRD was uploaded and validated but has not run: Codabench executes that competition on volunteered compute workers, none were available, and the queue is first-come-first-served, so providing a machine would not guarantee our own submission is the one it picks up. No EB-NeRD screenshot exists and producing one is not within our control.

on EB-NeRD where the columns exist and approximately nothing on the MIND leaderboard where they do not.

The offline prediction held. We projected 0.6460 for this exact 11-feature configuration on our own MIND test split and the leaderboard returned 0.6473, +0.0013 apart on a 2.37M-impression set we have no labels for. That is the strongest available evidence that the streaming submission path preserves the offline scoring semantics, which three abandoned implementations and a silent NaN mismatch had put in doubt.

One caution on the MRR column. MIND’s scorer averages reciprocal rank over every clicked article while the common definition credits only the first, so the “all clicks” figure is the one to compare here (Section 5).

5 Observations that generalise

A feature’s isolated score is not its contribution. Impression size, `n_cands`, is identical for every candidate in an impression, so alone it scores exactly 0.5000, yet removing it costs −0.0066 test AUC, the second largest loss in the grid: it helps only through interactions, by telling the model how crowded the impression

is. The reverse also happens, with freshness carrying high LightGBM split gain but removing it changing almost nothing, because popularity already holds the same information. We therefore quote ablation deltas, never importance.

Leave-one-out deltas do not add up. In the 17-feature grid, removing `pop_causal` alone or `user_read` alone makes the model slightly better on val. Removing both is significantly *worse* on both splits (−0.0005 val, −0.0007 test), and keeping only the features with significant positive deltas loses 0.0240 test AUC. Correlated features cover for each other, so a single-feature ablation answers “what does this add given the rest” and not “what should we keep”. We ship all features.

An underpowered result is not a null result. Our stage-one ablations first ran on the 10× smaller EB-NeRD demo set, where two intervals spanned zero. At full scale they resolved in opposite directions: dropping the abstract from BM25 went from +0.0009 [−0.0062, +0.0081] to a real +0.0025 [+0.0002, +0.0047], while Danish stemming stayed flat within ±0.0004, at intervals tight enough to exclude the +22 to +36% recall gain Assignment 1 had claimed. We now re-run any interval spanning zero at larger scale before recording a ver-

dict.

A check must test the artifact, not a proxy for it. We found **six** checks on this project that passed while testing nothing, the last of them an injection test that two functions sharing a name had silently deleted, so it ran zero times from the day it was written. Reading the file cannot catch that; counting collected tests can, and **12 collected against 15 written** is what exposed it. Every test must now fail on an injected violation before it counts, and how many actually run is part of what we check. All six, and the related case of a guard that fired on a comment rather than on code, are in Appendix C.

When a number fails on exactly one dataset, check the definition before the code. Three of our metrics turned out to carry two definitions each, and every one reproduced on one dataset and not the other, because the definitions diverge only when impressions carry several clicks (28.8% on MIND against 0.5% on EB-NeRD). The three are in Appendix D; one of them is how we found the leak.

6 Limitations and open questions

- **Parts of Q1 are not modelled.** No within-session features (EB-NeRD’s session id is unused) and no position-bias correction. Time decay on the history was neutral to harmful at every setting in Assignment 1 and was not revisited inside the ranker.
- **The baseline covers one dataset.** NRMS was reproduced and beaten on EB-NeRD only. The history length of 100 our sweep favours was not tried in NRMS, whose cost grows quadratically with it.
- **The final model may be under-trained.** Its best iteration is 399 of a 400-round cap, so it was still improving when training stopped; the reported gain is more likely understated than inflated.
- **Our strongest claims are about EB-NeRD.** MIND’s gain is a small fraction of it, a property of the data rather than the method, but one dataset carries most of the

evidence.

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A Background: the methods this note uses

Brief definitions, moved out of the body so the body can stay results-only.

BM25 [4, 5] is a lexical retrieval score. It rewards an article for sharing words with the query, weights each word by how rare it is across the collection, and saturates the reward so a word repeated ten times does not count ten times. Our query is the concatenated titles of the user’s last N clicks.

LambdaRank [9] is the training objective of the re-ranker. Instead of predicting a click probability per candidate, it looks at *pairs* of candidates inside one impression and weights each pair by how much swapping them would change nDCG. Training therefore optimises the ordering that the metric actually measures. Our alternative, a plain binary objective, is priced in Table 4.

Per-impression AUC is the probability that a randomly chosen clicked candidate outranks a randomly chosen unclicked one, computed *inside* a single impression and then averaged over impressions that contain both. The alternative, pooling every candidate from every impression into one ranking, measures something else: it rewards a feature for separating busy impressions from quiet ones even when that feature cannot reorder anything within an impression. The gap is large in practice. Article freshness reads **0.5861 pooled against 0.5087 per impression**, so pooling would credit it with real skill it does not have.

The paired bootstrap [10] is how every interval in this note is produced. We resample impressions with replacement 1,000 times and recompute the metric on each resample. It is *paired* because both systems are scored on the *same* resampled impressions, so the variance that comes from some impressions being intrinsically easier cancels between them. An unpaired interval on two systems that rank the same impressions is dominated by that shared variance and is far too wide to resolve the differences we care about. A gain counts only when its interval

excludes zero.

NRMS [3] is the neural baseline. It encodes each article into a vector, then builds a user representation by attention over that user’s clicked articles, so the model learns which past clicks matter for the current candidate. It reads article content and click order only, which is exactly why it cannot see the popularity signal our re-ranker leans on.

B The engagement leak: how we found it

EB-NeRD ships its history in two blocks, each describing clicks from before *that block* begins. Our feature builder merged both and let the later block overwrite the earlier for 11,658 users. Our val split is carved out of the earlier block, so val impressions were described by history drawn from their own future.

We found it by chasing a metric discrepancy, not with a test. Two implementations of the same user vector scored 0.6473 and 0.5506. Making one reproduce both numbers with only the history source changed confirmed the cause and priced it: a +0.0967 inflation on that feature. The fix reads one block per split, and a new test that fails on injection covers it.

Two consequences show the leak was real and the fix was right. The read-time-weighted user vector’s apparent +0.0085 contribution fell to +0.0002, not significant, so its entire value had been the leak. And test AUC *rose* from 0.7429 to 0.7461, because test features were always clean and only the model, now trained without future information, changed.

C Checks that checked nothing

Six checks on this project passed while testing nothing. They are listed because the failure mode is the same each time, an assertion that ran against the wrong object, and because it is the reason every leakage assertion here is now verified by injection before it is trusted.

1. A MIND leakage test filtered every row away before asserting. Its history column is a *null*

split	impressions	matched to the later history block	history at or after the impression
train	168,522	144,087	143,998 (99.9%)
val	64,365	61,026	60,901 (99.8%)
test	244,647	244,647	0 (0.0%)

Table 12: Future-click contamination in the engagement features before the fix (Section 2.5). Test was clean because it is drawn from the later block itself.

list of dtype `List(Datetime)`, not an empty list, so `.list.len()` returned null, a filter for `= 0` matched nothing, and the test reported green over 95,071 rows while checking none of them.

2. The anti-gaming guard scanned zero files after the Assignment-1 port moved everything under `src/`. It globbed directories that had stopped existing. It now asserts a minimum file count so an empty scan fails.
3. An install check trusted `pip`’s exit code while `import tensorflow` had been broken the whole time. It now imports and multiplies two matrices.
4. A job monitor matched its own command line in the process table and reported the job alive after it had died.
5. The leakage tests covered `pop_causal` only, so the engagement arrays the model also reads were never checked. That is what let the block-merge leak survive (Appendix B).
6. **An injection test deleted by its own name.** Two test functions in one file shared a name, so the second definition replaced the first and the popularity injection check ran **zero times** from the day it was written. Reading the file cannot catch this. Counting collected tests can: **12 collected against 15 written** is what exposed it.

A guard that fires on prose teaches you to delete the prose. The same anti-gaming test searched source as raw text and failed on `src/rerank/q9.py`, whose only mention of the lifetime columns is a comment recording that MIND has none of them. Allow-listing the file would have passed the suite while removing the guard from a module that never read those columns. It now parses the source and inspects identifiers and string literals in

executable code, so a comment costs nothing and `pl.col("total_inviews")` still fails. It remains a static scan: a column name assembled at runtime would slip past, and closing that needs a runtime check on what the scorer reads, which we did not build.

D Metric names that hid two definitions

Three times a number failed to reproduce and the cause was the definition, not the code. In each case it reproduced on one dataset and not the other, because these definitions only diverge when impressions carry several clicks, 28.8% on MIND against 0.5% on EB-NeRD. **When a number fails on exactly one dataset, check the definition first.**

- **Hit rate labelled recall.** Our retrieval stage printed the share of impressions with at least one clicked article retrieved, under the name for the share of clicked articles retrieved. On MIND test they read 0.0333 and 0.0226, and reconcile exactly: $0.0333 \times 0.6775 = 0.0226$.
- **Two MRRs.** MIND’s official scorer averages reciprocal rank over *every* clicked article; the common definition credits only the first. For one unchanged ranking they give 0.2693 and 0.3108, with AUC and nDCG identical. Our harness reports both and the “all clicks” column is the one comparable to the leaderboard.
- **Two user vectors.** Two implementations of the same feature read different history files and scored 0.6473 and 0.5506. That discrepancy is what uncovered the leak in Appendix B.

E Full ablation grids

Each arm removes one feature and retrains the same model; `stage1_only` keeps only the two retrieval scores and `pop_only` only click popularity. A positive “full – arm” means the removed feature helped. Bold marks intervals excluding zero. These grids were run before the exposure features were added (17 features on EB-NeRD, 12 on MIND), which is why their full-model AUC sits below the shipped system’s.

F Beyond-accuracy metrics

Coverage carries no interval, and our own failed attempt at one is the reason. It counts *distinct* articles rather than averaging over impressions, so it is not a mean and the bootstrap does not apply. A resample draws only about 63% of the impressions, and every article that only the missing impressions would have surfaced simply vanishes, so every resample undercounts. The bias is visible in our numbers: coverage on EB-NeRD test is **0.2067** for BM25 while the resampled spread is **[0.1875, 0.1920]**, an “interval” that excludes the value it is meant to describe. We report the point estimate and keep the spread only as a diagnostic. Diversity and novelty are means over impressions and do take intervals.

G Scaling measurements

H Isolated-feature sweeps

Per-impression AUC of one feature used alone as the score, on val. An isolated score cannot be absorbed by a correlated neighbour, which makes it a clean screen, but a winner here is only a candidate: the popularity window gained +0.0447 in isolation and +0.0034 val inside the ranker.

I Validation-split evaluation

J Reproducing the numbers

Every table in this note is generated from the results files by

`docs/design-note/make_tables.py`; every measured number also has a row in `docs/FACTS.md` or `docs/ABLATIONS.md` with the command that produced it.

K Leaderboard screenshots

MIND, Codabench competition 13967, submission 932812, user `namsyn`, rank 54, scored 2026-09-19 09:23: AUC 0.6473, MRR 0.3174, nDCG@5 0.3448, nDCG@10 0.4010. The image is in the repository at `docs/report/screenshots/` rather than reproduced here.

EB-NeRD, Codabench competition 2469: **no screenshot, and not for want of submitting.** The prediction file was built, validated against all five structural checks, scored on a cluster node in 3 h 06 m and uploaded. It has not run. Codabench executes that competition on volunteered compute workers, none were available, and the queue is first-come-first-served, so making a machine available would not guarantee our own submission is the one it picks up. Producing this screenshot is not within our control.

Nothing in this report depends on it: every EB-NeRD number quoted here is measured on our own labelled test split, never on the leaderboard. The single thing we cannot state is an EB-NeRD leaderboard rank, and we do not state one.

arm	val AUC	full – arm, val [95% CI]	full – arm, test [95% CI]
full model	0.7636		test AUC 0.7653
stage1_only	0.5498	+0.2138 [+0.2110, +0.2168]	+0.2223 [+0.2209, +0.2238]
pop_only	0.7348	+0.0288 [+0.0275, +0.0301]	+0.0360 [+0.0352, +0.0367]
minus_cat_match	0.7574	+0.0062 [+0.0053, +0.0071]	+0.0069 [+0.0064, +0.0074]
minus_n_cands	0.7580	+0.0056 [+0.0050, +0.0063]	+0.0066 [+0.0062, +0.0069]
minus_bm25	0.7625	+0.0011 [+0.0006, +0.0017]	+0.0004 [+0.0001, +0.0007]
minus_pop_rank	0.7629	+0.0007 [+0.0000, +0.0013]	+0.0003 [+0.0000, +0.0006]
minus_hist_len	0.7631	+0.0005 [+0.0000, +0.0010]	+0.0005 [+0.0002, +0.0007]
minus_pop_velocity	0.7632	+0.0004 [–0.0001, +0.0008]	–0.0002 [–0.0004, +0.0001]
minus_user_scroll	0.7632	+0.0003 [–0.0001, +0.0008]	–0.0003 [–0.0006, –0.0001]
minus_age_hours	0.7632	+0.0003 [–0.0001, +0.0008]	+0.0004 [+0.0002, +0.0006]
minus_engage_sim	0.7633	+0.0003 [–0.0003, +0.0010]	+0.0005 [+0.0002, +0.0008]
minus_emb	0.7637	–0.0001 [–0.0006, +0.0004]	–0.0003 [–0.0006, –0.0001]
minus_emb_rank	0.7637	–0.0001 [–0.0006, +0.0003]	–0.0001 [–0.0003, +0.0002]
minus_pop_24h	0.7638	–0.0003 [–0.0008, +0.0003]	–0.0001 [–0.0004, +0.0002]
minus_ent_overlap	0.7639	–0.0003 [–0.0008, +0.0002]	–0.0006 [–0.0009, –0.0004]
minus_pop_rel_max	0.7639	–0.0003 [–0.0009, +0.0003]	+0.0002 [–0.0002, +0.0005]
minus_recency	0.7639	–0.0004 [–0.0008, +0.0001]	–0.0005 [–0.0008, –0.0003]
minus_user_read	0.7641	–0.0005 [–0.0010, –0.0000]	–0.0006 [–0.0008, –0.0003]
minus_pop_causal	0.7644	–0.0008 [–0.0013, –0.0003]	–0.0006 [–0.0009, –0.0003]

Table 13: EB-NeRD small, 17-feature model, 64,365 val and 244,647 test impressions.

```

make venv && make download
python -m src.pipeline.split ebnerd_small          # one dataset per process
python -m src.retrieval.bm25 ebnerd_small
python -m src.retrieval.embeddings ebnerd_small
python -m src.retrieval.fuse ebnerd_small
python -m src.features.build ebnerd_small
python -m src.rerank.train ebnerd_small --tag final
python -m src.eval.run ebnerd_small --rerank-tag final
python -m src.eval.compare_reranks ebnerd_small ladder17 final
python -m src.eval.bench ebnerd_small --requests 2000 \
--scale-points 0.1 0.25 0.5 1.0 --model-tag final
python -m src.eval.pool_size ebnerd_small
python -m pytest tests/
python -m src.submission.validate ebnerd submissions/ebnerd_predictions.zip

```

arm	val AUC	full – arm, val [95% CI]	full – arm, test [95% CI]
full model	0.6539		test AUC 0.6666
pop_only	0.5475	+0.1064 [+0.1040, +0.1089]	+0.0957 [+0.0935, +0.0978]
stage1_only	0.6372	+0.0167 [+0.0147, +0.0187]	+0.0293 [+0.0274, +0.0313]
minus_emb	0.6497	+0.0042 [+0.0033, +0.0052]	+0.0010 [+0.0001, +0.0018]
minus_cat_match	0.6502	+0.0037 [+0.0028, +0.0047]	+0.0070 [+0.0062, +0.0079]
minus_bm25	0.6517	+0.0022 [+0.0013, +0.0031]	+0.0027 [+0.0020, +0.0035]
minus_hist_len	0.6530	+0.0008 [+0.0002, +0.0016]	–0.0001 [–0.0007, +0.0006]
minus_pop_causal	0.6531	+0.0008 [+0.0001, +0.0014]	–0.0009 [–0.0016, –0.0003]
minus_pop_velocity	0.6534	+0.0005 [–0.0003, +0.0013]	+0.0000 [–0.0007, +0.0007]
minus_n_cands	0.6537	+0.0001 [–0.0007, +0.0011]	+0.0009 [+0.0002, +0.0017]
minus_ent_overlap	0.6538	+0.0001 [–0.0006, +0.0007]	–0.0005 [–0.0012, +0.0001]
minus_pop_rank	0.6540	–0.0001 [–0.0008, +0.0006]	+0.0003 [–0.0004, +0.0010]
minus_pop_rel_max	0.6541	–0.0002 [–0.0010, +0.0007]	+0.0001 [–0.0007, +0.0009]
minus_pop_24h	0.6542	–0.0003 [–0.0010, +0.0004]	+0.0003 [–0.0003, +0.0010]
minus_emb_rank	0.6542	–0.0003 [–0.0010, +0.0004]	–0.0007 [–0.0014, –0.0001]

Table 14: MIND small, 12-feature model, 61,894 val and 73,152 test impressions.

system	diversity	novelty (bits)	coverage
<i>EB-NeRD small, test</i>			
BM25 (stage one)	0.7968 [0.7961, 0.7974]	16.966 [16.964, 16.969]	0.2067
embeddings (stage one)	0.7803 [0.7796, 0.7809]	17.007 [17.004, 17.010]	0.2050
fused (stage one)	0.7820 [0.7814, 0.7826]	16.999 [16.996, 17.002]	0.2051
two-stage re-ranker	0.7901 [0.7895, 0.7907]	17.084 [17.082, 17.086]	0.2026
<i>MIND small, test</i>			
BM25 (stage one)	0.8428 [0.8416, 0.8439]	16.927 [16.924, 16.929]	0.0536
embeddings (stage one)	0.8215 [0.8202, 0.8227]	16.926 [16.923, 16.929]	0.0523
fused (stage one)	0.8182 [0.8169, 0.8195]	16.925 [16.922, 16.927]	0.0529
two-stage re-ranker	0.8008 [0.7996, 0.8021]	17.003 [17.001, 17.005]	0.0494

Table 15: Top-10 beyond-accuracy metrics, test split. Diversity is the share of top-10 pairs from different categories; novelty is mean self-information in bits, so higher is less obvious; coverage is the share of the catalogue that ever reaches a top 10.

corpus	articles	FAISS MiB	BM25 p50	exact kNN p50	rerank p50	total p50	total p99
10%	2,074	6.1	0.24	0.79	2.33	3.94	7.28
25%	5,184	15.2	0.40	1.66	2.20	4.99	8.42
50%	10,369	30.4	0.69	3.13	2.35	7.00	10.22
100%	20,738	60.8	3.93	21.11	17.66	46.87	109.00

Table 16: Latency against corpus size in milliseconds, the numbers behind Figure 2. Same machine and same model throughout.

popularity window	EB-NeRD	MIND	min. scroll depth	AUC	history kept
1 h	0.7082	0.5622	0%	0.5545	100.0%
6 h	0.7349	0.5475	10%	0.5521	88.5%
24 h	0.7254	0.5516	25%	0.5559	78.5%
72 h	0.7045	0.5506	50%	0.5590	58.4%
168 h	0.6902	0.5502	75%	0.5606	46.7%
unbounded	0.6902	0.5502	90%	0.5609	41.1%

recency half-life	1 h	3 h	6 h	12 h	24 h	48 h	96 h	168 h
EB-NeRD AUC	0.5088	0.5087	0.5087	0.5088	0.5087	0.5088	0.5088	0.5088
history clicks	1	5	10	20	50	100		
uniform user vector	0.5221	0.5368	0.5416	0.5483	0.5528	0.5545		
read-time weighted	0.5221	0.5362	0.5434	0.5495	0.5554	0.5570		

Table 17: Top left: the click-popularity window. Top right: dropping history clicks with low scroll depth (EB-NeRD, 100 clicks). Middle: the recency half-life, flat to four decimals over a $168\times$ range. Bottom: history length and read-time weighting of the user vector on clean history.

system	AUC	MRR	MRR, all clicks	nDCG@5	nDCG@10
<i>EB-NeRD small, val</i>					
BM25 (stage one)	0.5205 [0.5180, 0.5231]	0.3418 [0.3395, 0.3440]	0.3414 [0.3391, 0.3436]	0.3794 [0.3767, 0.3821]	0.4607 [0.4586, 0.4629]
embeddings (stage one)	0.5506 [0.5482, 0.5530]	0.3594 [0.3572, 0.3617]	0.3589 [0.3567, 0.3612]	0.4017 [0.3991, 0.4044]	0.4778 [0.4757, 0.4801]
fused (stage one)	0.5528 [0.5503, 0.5553]	0.3628 [0.3604, 0.3652]	0.3623 [0.3600, 0.3647]	0.4043 [0.4016, 0.4071]	0.4807 [0.4785, 0.4830]
two-stage re-ranker	0.7885 [0.7867, 0.7906]	0.5729 [0.5703, 0.5755]	0.5724 [0.5698, 0.5750]	0.6411 [0.6387, 0.6437]	0.6715 [0.6695, 0.6736]
fused + lifetime popularity*	0.5784 [0.5760, 0.5808]	0.3707 [0.3684, 0.3731]	0.3703 [0.3679, 0.3726]	0.4184 [0.4159, 0.4210]	0.4910 [0.4889, 0.4932]
<i>MIND small, val</i>					
BM25 (stage one)	0.5840 [0.5816, 0.5864]	0.3204 [0.3176, 0.3232]	0.2777 [0.2753, 0.2803]	0.2924 [0.2895, 0.2954]	0.3494 [0.3467, 0.3522]
embeddings (stage one)	0.6356 [0.6334, 0.6379]	0.3422 [0.3396, 0.3449]	0.2988 [0.2964, 0.3012]	0.3202 [0.3175, 0.3231]	0.3789 [0.3763, 0.3816]
fused (stage one)	0.6390 [0.6368, 0.6412]	0.3478 [0.3450, 0.3504]	0.3031 [0.3007, 0.3056]	0.3248 [0.3220, 0.3275]	0.3836 [0.3811, 0.3863]
two-stage re-ranker	0.6997 [0.6976, 0.7018]	0.3977 [0.3950, 0.4005]	0.3422 [0.3397, 0.3448]	0.3729 [0.3701, 0.3759]	0.4313 [0.4287, 0.4339]

Table 18: All accuracy metrics on val with 95% intervals.